PROBABILITY | LECTURE 03

Birthday Problem and Probability Properties

Edited notes based on the handwritten pages and the accompanying lecture transcript

1 The Birthday Problem

1.1 Question and assumptions

Suppose a group contains k *distinct, labeled people*. What is the probability that *at least two* people share a birthday? We are not asking whether a particular pair matches; *any* pair is sufficient.

To make the problem mathematically tractable, assume:

1. **365 possible birthdays:** ignore February 29.
2. **Uniformity:** each of the 365 dates is equally likely for every person.
3. **Independence:** one person's birthday does not affect another person's birthday.

These are simplifying modeling assumptions. Actual birthdays are not perfectly uniform across the year, and birthdays of relatives such as twins need not be independent.

1.2 The pigeonhole principle: more than 365 people

Picture the 365 days as boxes and each person as a dot in the box corresponding to their birthday:



If there are more than 365 people for only 365 birthday boxes, some box must contain at least two people. This is the **pigeonhole principle**:

$$\boxed{\mathbb{P}(\text{at least one birthday match}) = 1 \qquad \text{if } k > 365.}$$

The same idea appears in computer science: when objects are mapped to a limited number of storage locations, two objects may map to the same location, producing a *collision* (as in hashing).

1.3 Calculating the probability with at most 365 people

It is easier to calculate the **complementary event**: *no two people share a birthday.*

Because all birthday assignments are equally likely and independent, there are 365^k total assignments. To obtain *no match*:

- Person 1 can have any of 365 birthdays.
- Person 2 must avoid that date: 364 choices.
- Person 3 must avoid the first two dates: 363 choices.

- Person k has $365 - k + 1$ choices.

Thus, with exactly k factors in the numerator,

$$\begin{aligned}\mathbb{P}(\text{no match}) &= \frac{365 \cdot 364 \cdot 363 \cdots (365-k+1)}{365^k} \\ &= \prod_{i=0}^{k-1} \frac{365-i}{365} \\ &= \frac{365!}{(365-k)!\,365^k}.\end{aligned}$$

Birthday-match formula

$$\boxed{\mathbb{P}(\text{at least one match}) = \begin{cases} 1 - \dfrac{365!}{(365-k)!\,365^k}, & 1 \le k \le 365, \\ 1, & k > 365. \end{cases}}$$

For $k = 1$, the no-match probability is $365/365 = 1$, as expected. The final factor $365 - k + 1$ is important: omitting the $+1$ is an *off-by-one error*.

1.4 Numerical results

People k	Probability of a match	Interpretation
22	47.57%	Below 50%
23	50.73%	First value above 50%
50	97.04%	Very likely
100	99.999969%	Almost certain

Key conclusion: Only $\boxed{23}$ people are needed to make the probability of at least one shared birthday greater than 50%.

1.5 Why 23 people are enough

The crucial quantity is not just the number of people k, but the number of *different pairs* among them:

$$\binom{k}{2} = \frac{k(k-1)}{2}.$$

For $k = 23$,

$$\boxed{\binom{23}{2} = \frac{23 \cdot 22}{2} = 253 \text{ possible pairs.}}$$

Each pair is an opportunity for a birthday coincidence. There are 253 possible pairs even though there are only 23 people. *These pairwise matching events are not mutually independent*, so this counting argument is intuition, not a replacement for the exact formula above.

Related variation. If two birthdays are counted as a match when they are the same *or one calendar day apart*, the lecture states that 14 people already give a probability exceeding 50%. This version is harder to calculate exactly because the dates excluded by different people can overlap; the simple falling-product argument does not directly apply.

2 The Non-Naive Definition of Probability

Let $\mathcal{S}$ be a sample space, and let events be subsets of $\mathcal{S}$. A probability function assigns a number $\mathbb{P}(A)$ to each event A.

Range: for every event A,

$$0 \leq \mathbb{P}(A) \leq 1.$$

The lecture treats this nonnegativity/range requirement as part of what is meant by a probability function (sometimes called an "axiom zero").

2.1 Axioms

Axiom 1: Impossible and certain events

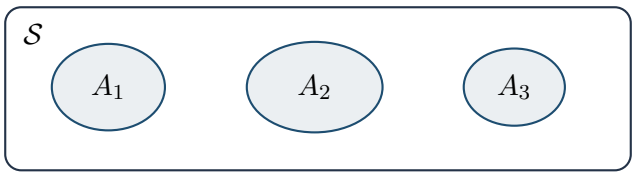
$$\boxed{\mathbb{P}(\varnothing) = 0, \qquad \mathbb{P}(\mathcal{S}) = 1.}$$

The empty event can never occur; the full sample space always occurs.

Axiom 2: Countable additivity for disjoint events
If $A_1, A_2, \ldots$ are **pairwise disjoint** (that is, $A_i \cap A_j = \varnothing$ when $i \neq j$), then

$$\boxed{\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).}$$

In particular, if $A \cap B = \varnothing$, then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.



Intuition: Think of the total area of $\mathcal{S}$ as 1. If the regions do not overlap, the area of their union is just the sum of their areas. These abstract axioms, associated with Kolmogorov's mathematical formulation of probability, let us prove results without first settling philosophical interpretations of probability.

3 Three Fundamental Properties

3.1 Property 1: Complement rule

$$\boxed{\mathbb{P}(A^c) = 1 - \mathbb{P}(A).}$$

Proof. Because $A \cup A^c = \mathcal{S}$ and $A \cap A^c = \varnothing$, additivity gives

$$\begin{aligned} 1 &= \mathbb{P}(\mathcal{S}) \\ &= \mathbb{P}(A \cup A^c) \\ &= \mathbb{P}(A) + \mathbb{P}(A^c). \end{aligned}$$

Rearrange to obtain $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. □

3.2 Property 2: Monotonicity

If $A \subseteq B$, then

$$\boxed{\mathbb{P}(A) \leq \mathbb{P}(B).}$$

In words: if event A occurs, then event B necessarily occurs, so B cannot be less likely than A.

Proof. When $A \subseteq B$, split B into two disjoint pieces:

$$B = A \cup (B \cap A^{\mathrm{c}}).$$

Therefore,

$$\begin{aligned}\mathbb{P}(B) &= \mathbb{P}(A) + \mathbb{P}(B \cap A^{\mathrm{c}}) \\ &\geq \mathbb{P}(A),\end{aligned}$$

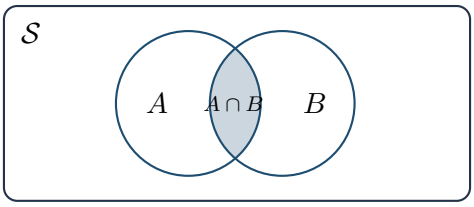
where the inequality follows from nonnegativity. □

3.3 Property 3: Addition rule for two events

For *any* events A and B, not necessarily disjoint,

$$\boxed{\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).}$$

Intuition. $\mathbb{P}(A) + \mathbb{P}(B)$ counts the overlap $A \cap B$ twice. Subtract it once to count every outcome in the union exactly once.



Proof by "disjointification." Rewrite the union as a union of disjoint parts:

$$A \cup B = A \cup (B \cap A^{\mathrm{c}}).$$

Hence

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B \cap A^{\mathrm{c}}). \tag{1}$$

Now split B into the disjoint events $A \cap B$ and $A^{\mathrm{c}} \cap B$:

$$B = (A \cap B) \cup (A^{\mathrm{c}} \cap B).$$

By additivity,

$$\mathbb{P}(B) = \mathbb{P}(A \cap B) + \mathbb{P}(A^{\mathrm{c}} \cap B),$$

so

$$\mathbb{P}(B \cap A^{\mathrm{c}}) = \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Substitute into (1) to obtain the addition rule. □

4 The Inclusion–Exclusion Principle

The addition rule generalizes to unions of three or more events. We first *include* individual probabilities, *exclude* overlap terms, and continue alternating to correct the counts.

4.1 Three events

For three events A, B, C,

$$\boxed{\begin{aligned}\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) \\ - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) \\ + \mathbb{P}(A \cap B \cap C).\end{aligned}}$$

Why the triple intersection is added back. Any point in $A \cap B \cap C$ is initially counted three times, then removed three times with the pairwise subtractions. It is therefore counted zero times until we add it once more: $3 - 3 + 1 = 1$.

4.2 General form for multiple events

For events $A_1, \ldots, A_n$,

$$\begin{aligned}\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} \mathbb{P}(A_i) - \sum_{1 \le i < j \le n} \mathbb{P}(A_i \cap A_j) \\ + \sum_{1 \le i < j < k \le n} \mathbb{P}(A_i \cap A_j \cap A_k) - \cdots \\ + (-1)^{n+1} \mathbb{P}\left(\bigcap_{i=1}^{n} A_i\right).\end{aligned}$$

Equivalently, a compact notation for the same formula is

$$\boxed{\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{\varnothing \neq I \subseteq \{1,\ldots,n\}} (-1)^{|I|+1} \mathbb{P}\left(\bigcap_{i \in I} A_i\right).}$$

Here I runs over all nonempty index sets; $|I|$ denotes the number of indices in I. The signs alternate: single intersections are added, pairwise intersections subtracted, triple intersections added, and so on. The formula can be proved by repeatedly applying the two-event rule or by induction.

5 de Montmort's Matching Problem (1713)

5.1 Description of the game

There are n different cards labeled $1, 2, \ldots, n$. Shuffle them uniformly at random and reveal them in order. You **win** if the number printed on at least one card agrees with that card's *position* in the shuffled deck. For example, the seventh position contains card number 7.

This is the **matching problem**, also known as **de Montmort's problem**. In the language of permutations, it asks for the probability that a uniformly random permutation has *at least one fixed point*.

Define the events

$$A_j = \{\text{the card in position } j \text{ is labeled } j\}, \qquad j = 1, \ldots, n.$$

We want

$$\mathbb{P}(\text{at least one match}) = \mathbb{P}(A_1 \cup A_2 \cup \cdots \cup A_n).$$

5.2 Finding the intersection probabilities

There are $n!$ equally likely orderings of the deck.

One specified match. If position j must contain card j, the other $n-1$ cards may be arranged in $(n-1)!$ ways, so

$$\boxed{\mathbb{P}(A_j) = \frac{(n-1)!}{n!} = \frac{1}{n}.}$$

This does not depend on j, which is the symmetry that makes the problem manageable.

Two specified matches. If positions 1 and 2 contain cards 1 and 2, the remaining $n-2$ cards can be arranged arbitrarily:

$$\boxed{\mathbb{P}(A_1 \cap A_2) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}.}$$

The same probability applies to any two distinct specified positions.

k specified matches. More generally, if $i_1, \dots, i_k$ are distinct, fixing those k positions leaves $(n-k)!$ arrangements:

$$\boxed{\mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}) = \frac{(n-k)!}{n!}.}$$

Again, this depends on k, not on which positions are chosen.

5.3 Applying inclusion–exclusion

There are $\binom{n}{k}$ ways to choose which k positions match. Thus the total contribution from all intersections of k distinct events is

$$\binom{n}{k}\frac{(n-k)!}{n!} = \frac{n!}{k!(n-k)!}\frac{(n-k)!}{n!} = \frac{1}{k!}.$$

By the alternating signs of inclusion–exclusion,

$$\begin{aligned}\mathbb{P}(\text{at least one match}) &= n \cdot \frac{1}{n} - \binom{n}{2}\frac{(n-2)!}{n!} + \binom{n}{3}\frac{(n-3)!}{n!} - \cdots \\ &= 1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \cdots + \frac{(-1)^{n+1}}{n!}.\end{aligned}$$

Exact probability of at least one match

$$\boxed{\mathbb{P}(\text{at least one match}) = \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k!}.}$$

5.4 Connection with the exponential function

Recall the Taylor series

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

Putting $x = -1$ gives

$$e^{-1} = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \cdots .$$

Therefore, as the number of cards grows,

$$\boxed{\mathbb{P}(\text{at least one match}) \longrightarrow 1 - \frac{1}{e} \approx 0.63212056 \approx 63.2\%.}$$

The complementary event is *no fixed points*, also called a *derangement*. Its probability is

$$\mathbb{P}(\text{no match}) = \sum_{k=0}^{n} \frac{(-1)^k}{k!} \longrightarrow \frac{1}{e} \approx 36.8\%.$$

The appearance of e comes from the factorial-denominator alternating series, not from any special property of playing cards.

Key Takeaways

- **Complement first:** In the birthday problem, calculate no match and subtract from 1.
- **Think in pairs:** k people create $\binom{k}{2}$ opportunities for a birthday match.
- **Disjointify:** Probability additivity applies directly only to disjoint events; split overlapping sets into disjoint parts to prove new rules.
- **Inclusion–exclusion:** Add individual event probabilities, subtract pairwise overlaps, add triple overlaps, and continue alternating.
- **Exploit symmetry:** In the card matching problem, every k-fold intersection has the same probability, leading to the $1/k!$ simplification.